

APPLIED STATISTICAL ANALYSIS & ESTIMATION REPORT

Detailed Solutions and Case Formulations across Five Industry Domains

1. Food Delivery: Estimation of Mean Delivery Duration in Chennai

Statistical Problem Formulation:

- **Target Population:** The entire set of all food orders fulfilled across Chennai by the delivery platform over the operational period ($N \rightarrow \infty$).
- **Observed Sample:** A simple random sample (SRS) of $n = 100$ completed delivery trips.
- **Sample Mean Observed ($\bar{x}$):** 32.0 minutes.

Population Parameter vs. Estimator Identification:

- **Population Parameter (μ):** The parameter under investigation is the true, unknown population mean delivery time (μ) across all historical and current deliveries in Chennai:

$$\mu = (1 / N) \sum_{i=1}^N X_i$$

- **Sample Estimator (X):** The arithmetic mean of the sample data serves as the point estimator:

$$X = (1 / n) \sum_{i=1}^n X_i$$

- **Calculated Point Estimate ($\hat{\mu}$):**

$$\hat{\mu} = \bar{x} = 32.0 \text{ minutes}$$

Theoretical Statistical Properties:

- **Unbiasedness ($E[X] = \mu$):** The expected value of the sample mean across all theoretically possible random samples of size 100 is identical to the true population mean μ .
- **Asymptotic Normality (Central Limit Theorem):** Because the sample size $n = 100 > 30$, the sampling distribution of X approximates a Gaussian normal distribution regardless of the underlying skewness in individual delivery times (e.g., peak-hour traffic delays).

Business & Operational Implications:

- **Service Level Agreement (SLA) Calibration:** The estimated 32-minute baseline enables algorithmic dispatch systems to quote realistic estimated times of arrival (ETAs) to consumers, reducing customer churn.
- **Fleet Routing Optimization:** Operations managers can benchmark specific micro-markets within Chennai (e.g., Anna Nagar, OMR, Velachery) against this 32-minute baseline to identify delivery clusters experiencing severe road congestion or kitchen delays.

2. E-Commerce: Post-Impression Advertising Conversion Rate

Statistical Problem Formulation:

- **Target Population:** All prospective unique visitors who view the promotional banner advertisement on the e-commerce platform.
- **Sample Size (n):** 800 randomly selected visitors who were exposed to the ad.
- **Observed Conversion Count (x):** 176 visitors who completed a transaction.

Point Estimation Calculations & Error Metrics:

- **Point Estimate of Population Proportion ($\hat{p}$):**

$$\hat{p} = x / n = 176 / 800 = 0.22 \text{ (22.0\%)}$$

- **Complementary Non-Conversion Proportion ($\hat{q}$):**

$$\hat{q} = 1 - \hat{p} = 1 - 0.22 = 0.78 \text{ (78.0\%)}$$

- **Estimated Standard Error of the Proportion ($SE_{\hat{p}}$):**

$$SE_{\hat{p}} = \sqrt{[\hat{p}(1 - \hat{p}) / n]} = \sqrt{[0.22 \times 0.78] / 800} = \sqrt{0.1716 / 800} \approx 0.01464 \text{ (1.464\%)}$$

- **95% Error Margin Baseline ($ME = 1.96 \times SE$):**

$$ME = 1.96 \times 0.01464 \approx 0.0287 \text{ (2.87\%)}$$

Marketing Analytics & Economic Impact:

- **Return on Ad Spend (ROAS):** A 22% conversion rate provides the exact input required for media-mix modeling. If the customer acquisition cost (CAC) derived from this 22% rate remains below the average gross margin per sale, the ad campaign is financially viable for scaling.
- **Attribution Modeling:** Digital marketing analysts can use this baseline to measure incremental lift against a control group of visitors who were not shown the promotional ad.

3. Banking: Frequentist Foundation of Credit-Card Expenditure Estimation

Statistical Problem Formulation:

- **Target Population:** The entire portfolio of active primary credit-card holders of the commercial bank.
- **Sample Size (n):** 150 randomly drawn account records.
- **Sample Mean Observed ($\bar{x}$):** ₹24,500.

Theoretical Foundations of the Frequentist Paradigm:

The frequentist approach approaches point estimation through the lens of objective, long-run relative frequency rather than subjective probability distributions assigned to parameters:

- **Deterministic, Fixed Parameter (μ):** Unlike Bayesian statistics (which treats parameters as random variables with prior distributions), the frequentist approach treats the true mean expenditure μ as a fixed, non-random, unknown state of nature.
- **The Sampling Distribution Concept:** If the bank repeatedly takes infinite hypothetical random samples of $n = 150$ credit-card holders from the exact same population, the observed sample mean $\bar{x}_k$ will vary from sample to sample according to a well-defined probability distribution:

$$X \sim N(\mu, \sigma^2 / n)$$

- **Unbiased Estimator Property:** Because $E[X] = \mu$, the estimator exhibits zero systematic error. The sample value of ₹24,500 represents a single realization from this perfectly centered sampling distribution.
- **Maximum Likelihood Estimator (MLE):** Under the standard assumption of continuous expenditure modeling, X is the exact maximum likelihood estimator that maximizes the likelihood function $L(\mu, \sigma^2 | X_1, \dots, X_n)$.

Banking Risk & Portfolio Management:

- **Credit Limit & Underwriting Strategy:** Establishing that the average active cardholder spends ₹24,500 monthly allows risk management to evaluate revolving debt exposures.
- **Interchange Fee Revenue Forecasting:** Finance desks can multiply $\hat{\mu} = ₹24,500$ across total active cards (N) and the standard interchange percentage (e.g., 1.5%) to estimate quarterly transaction fee revenues.

4. Healthcare / Streaming Platform: Daily Engagement Proportion

Statistical Problem Formulation:

- **Target Population:** All registered active subscribers on the video-on-demand (VOD) streaming platform.
- **Sample Size (n):** 500 randomly selected subscribers.
- **Success Criterion (x):** Users who stream at least one full movie daily ($x = 320$).

Mathematical Point Estimation & Dispersion:

- **Point Estimate of Proportion ($\hat{p}$):**

$$\hat{p} = x / n = 320 / 500 = 0.64 \text{ (64.0\%)}$$

- **Point Estimate Variance ($Var(\hat{p})$):**

$$Var(\hat{p}) = [\hat{p}(1 - \hat{p})] / n = (0.64 \times 0.36) / 500 = 0.2304 / 500 = 0.0004608$$

- **Standard Error of Proportion ($SE_{\hat{p}}$):**

$$SE_{\hat{p}} = \sqrt{0.0004608} \approx 0.02147 \text{ (2.15\%)}$$

Product Analytics & Infrastructure Interpretation:

- **Daily Active User (DAU) Ratio:** An estimate of 64% daily movie viewership represents high engagement "stickiness." It indicates that for any arbitrary group of 10,000 active platform accounts, approximately 6,400 will generate daily playback traffic.
- **Content Delivery Network (CDN) Provisioning:** Infrastructure engineering teams rely on this 64% metric to size edge-server caching capacity and reserve multi-gigabit bandwidth to prevent peak evening buffering.
- **Churn Prediction Modeling:** Users whose streaming frequencies fall significantly below this 64% point estimate can be algorithmically targeted with automated content recommendations before they cancel their subscriptions.

5. Manufacturing: 95% Confidence Interval for Smartphone Battery Performance

Statistical Formulation & Given Sample Data:

- **Target Population:** All smartphones assembled under the current production batch.
- **Sample Size (n):** 64 devices (Large sample size, $n \geq 30$, allowing normal distribution approximations via CLT).
- **Sample Mean Lifetime ($\bar{x}$):** 18.5 hours.
- **Sample Standard Deviation (s):** 2.0 hours.
- **Confidence Level ($1 - \alpha$):** $0.95 \Rightarrow \alpha = 0.05$.
- **Two-Tailed Critical Value ($z_{\alpha/2}$):** For $\alpha/2 = 0.025$, the standard normal critical value is $z_{0.025} = 1.96$.

Step-by-Step Interval Construction:

- **Standard Error of the Mean (SE):**

$$SE = s / \sqrt{n} = 2.0 / \sqrt{64} = 2.0 / 8 = 0.25 \text{ hours}$$

- **Margin of Error Calculation (ME):**

$$ME = z_{\alpha/2} \times SE = 1.96 \times 0.25 = 0.49 \text{ hours}$$

• **Construction of the Lower and Upper Confidence Limits:**

$$\text{Lower Limit (LL)} = \bar{x} - ME = 18.5 - 0.49 = 18.01 \text{ hours}$$

$$\text{Upper Limit (UL)} = \bar{x} + ME = 18.5 + 0.49 = 18.99 \text{ hours}$$

$$CI_{95\%} = [18.01 \text{ hours}, 18.99 \text{ hours}]$$

Comprehensive Statistical & Engineering Interpretation:

- **Formal Statistical Meaning:** If this sampling process were repeated 100 times under identical conditions, approximately 95 of the constructed confidence intervals would capture the true, unknown population mean battery life (μ).
- **Quality Assurance & Compliance Verdict:**
 - If the marketing claim or regulatory specification requires an average battery life of at least 18.0 hours, the manufacturing line satisfies compliance because the entire 95% confidence interval (18.01 h to 18.99 h) is strictly greater than the 18.0-hour threshold.
 - The narrow interval width ($2 \times 0.49 = 0.98 \text{ hours}$) confirms that the manufacturing process has high stability and low batch-to-batch operational variance.